Project Management Documentation

Monte Carlo Based Risk Analysis Engine

with Backtesting Framework

A Comprehensive Study in Stochastic Modeling,
Quantitative Risk Assessment,
and Software Project Planning

Submitted By

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Executive Summary

This document presents comprehensive project management analysis for developing a Monte Carlo Based Risk Analysis Engine with Backtesting Framework. The system implements Monte Carlo simulation using Geometric Brownian Motion (GBM) to model asset prices, calculates risk metrics including Value at Risk (VaR) and Conditional Value at Risk (CVaR), and validates predictions through historical backtesting.

The project follows rigorous software engineering methodologies including Work Breakdown Structure (WBS), critical path analysis, resource optimization, and schedule compression techniques. The document comprises four parts: WBS decomposition, detailed project planning with network diagrams and Gantt charts, resource allocation with histograms, and schedule compression analysis.

The baseline project duration is 231 days, which through strategic compression reduces to a more efficient timeline, providing adequate buffer for quality assurance and final deliverables.

1 Part 1: Work Breakdown Structure (WBS)

1.1 WBS Overview

The WBS provides hierarchical decomposition of project scope into manageable components. It serves to define scope, identify work packages, establish cost and schedule foundations, and facilitate stakeholder communication.

The project is structured into major development phases following a systematic software development lifecycle adapted for quantitative finance applications.

1.2 Major Project Phases

	Name	Duration	Start	Finish
	Project Kickoff & Scope Freeze	60 days?	**23/1/26 8:00 AM**	**16/4/26 5:00 PM**
	Project Kickoff & Scope Freeze	2 days	23/1/26 8:00 AM	26/1/26 5:00 PM
	Feasibility Analysis & Lit. Review (GBM, VaR, CVaR)	4 days	23/1/26 8:00 AM	28/1/26 5:00 PM
	Data Source Selection & Justification	1 day	29/1/26 8:00 AM	29/1/26 5:00 PM
	Version Control & Repo Initialization	1 day	23/1/26 8:00 AM	23/1/26 5:00 PM
	Mathematical Model Specification	10 days	**29/1/26 8:00 AM**	**11/2/26 5:00 PM**
	Data Preprocessing & Return Definition	9 days?	**12/2/26 8:00 AM**	**24/2/26 5:00 PM**
	VaR & CVaR Computation Module	10 days	**25/2/26 8:00 AM**	**10/3/26 5:00 PM**
	System Integration & Visualization	23 days	**25/2/26 8:00 AM**	**27/3/26 5:00 PM**
	Testing & Validation	4 days	**30/3/26 8:00 AM**	**2/4/26 5:00 PM**
	Final Reporting & Submission	10 days	**3/4/26 8:00 AM**	**16/4/26 5:00 PM**

Figure 1: Major Project Phases

1.2.1 1. Project Kickoff & Scope Freeze

Duration: 50 days

This initial phase establishes the project foundation by defining objectives, identifying stakeholders, and finalizing the scope. The scope is formally frozen to prevent scope creep throughout development.

Key activities include project charter development, stakeholder analysis, and preliminary feasibility assessment.

1.2.2 2. Feasibility Analysis & Literature Review

Duration: 4 days

A focused review of existing Monte Carlo methodologies in financial risk assessment, examining Geometric Brownian Motion (GBM) for asset price modeling, and surveying current VaR and CVaR calculation techniques used in industry.

Deliverables include feasibility report and literature review document covering stochastic modeling approaches.

1.2.3 3. Data Collection & Initialization

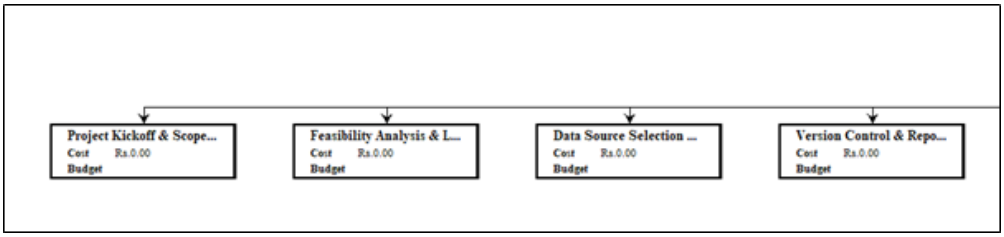
Duration: 1 day

Setting up data acquisition infrastructure and initializing connections to financial data sources. This includes configuring Yahoo Finance API access for historical OHLC (Open, High, Low, Close) price data.

1.2.4 4. Version Control & Repo Initialization

Duration: 1 day

Establishing Git repository structure, branching strategy, and collaborative development workflows. Setting up continuous integration pipelines and code review processes.



1.2.5 5. Mathematical Model Specification

Duration: 10 days

Formal specification of the mathematical foundation including GBM stochastic process, VaR and CVaR definitions, and discretization schemes for numerical simulation. This phase ensures mathematical rigor before implementation begins.

1.2.6 6. Mathematical Requirements Documentation

Duration: 4 days

Creating comprehensive documentation of all mathematical formulations, assumptions, and constraints. This includes parameter estimation methods and statistical validation criteria.

1.2.7 7. System Architecture Design

Duration: 2 days

Designing the modular pipeline architecture: Data Ingestion → Preprocessing → Monte Carlo Engine → Risk Metrics Calculation → Visualization. Establishing clear interfaces between components.

1.2.8 8. Module Interface Definition

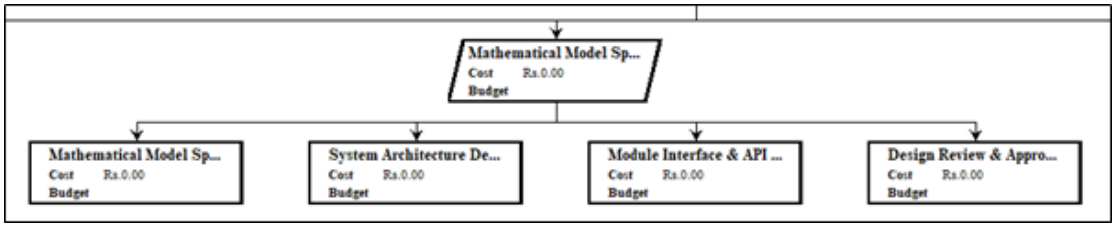
Duration: 15 days

Detailed specification of APIs, data formats, and communication protocols between system modules. Includes JSON schemas for data exchange and function signatures for all major components.

1.2.9 9. Design Review & Approval

Duration: 2 days

Formal review of all design documents with stakeholders. Obtaining approval before proceeding to implementation phase.



1.2.10 10. Data Preprocessing & Return Definition

Duration: 3 days

Implementing data cleaning, handling missing values, adjusting for stock splits and dividends, and computing logarithmic returns from historical price data.

1.2.11 11. Simplified Rolling-Window Implementation

Duration: 10 days

Developing the rolling window mechanism for parameter estimation (drift and volatility) using historical data. This enables dynamic model calibration over time.

1.2.12 12. Vectorization: Seeding & Optimization

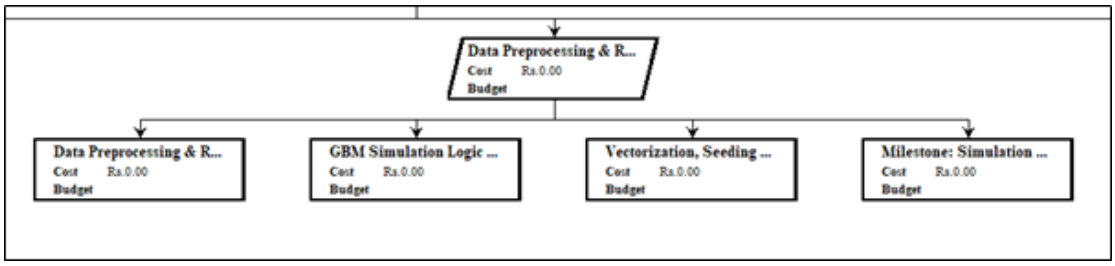
Duration: 2 days

Optimizing Monte Carlo simulation performance using NumPy vectorization, implementing efficient random number generation with proper seeding for reproducibility.

1.2.13 13. GBM Simulation Engine Ready

Duration: 0 days (Milestone)

Milestone marking completion of core Geometric Brownian Motion simulation engine capable of generating thousands of price path scenarios.



1.2.14 14. VaR & CVaR Computation Module

Duration: 10 days

Implementing risk metric extraction from simulated distributions. Calculating Value at Risk and Conditional Value at Risk at specified confidence levels (typically 95% and 99%).

1.2.15 15. Simplified Rolling-Window Backtesting

Duration: 4 days

Developing backtesting framework that validates VaR predictions against historical returns using rolling window approach. Tracks breach rates to verify model accuracy.

1.2.16 16. Breach Detection & Interpretation Logic

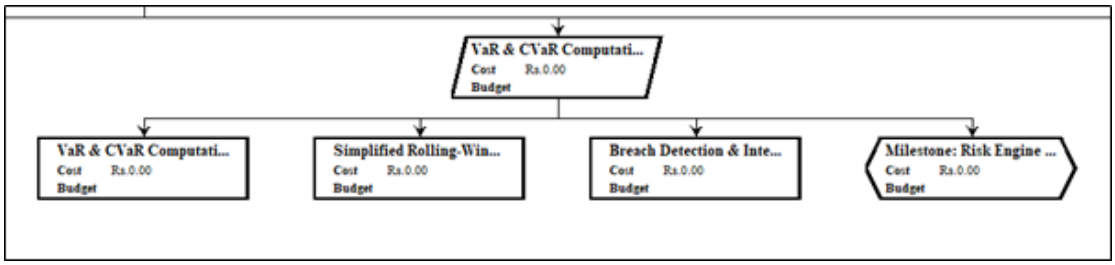
Duration: 2 days

Creating logic to identify when actual returns exceed predicted VaR thresholds and generating interpretable reports on model performance.

1.2.17 17. Milestone: Risk Engine Ready

Duration: 0 days (Milestone)

Major milestone indicating completion of fully functional risk analysis engine with backtesting capabilities.



1.2.18 18. System Integration (Simulation + Risk + Backtest)

Duration: 8 days

Integrating all components into cohesive end-to-end pipeline. Ensuring seamless data flow from ingestion through simulation to risk calculation and backtesting.

1.2.19 19. Risk Aggregation & Visualization

Duration: 8 days

Developing visualization dashboard using Matplotlib/Seaborn for probability distributions, sample paths, breach analysis, and statistical summaries.

1.2.20 20. Optional Demo Notebook (Non-critical)

Duration: 4 days

Creating Jupyter notebook demonstrations showcasing system capabilities with example use cases and explanatory commentary.

1.2.21 21. Performance & Acceleration Checks

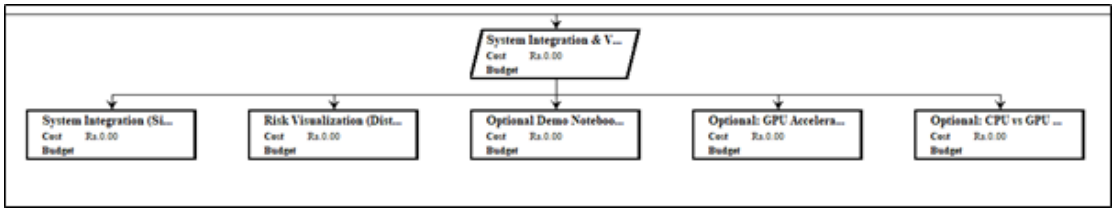
Duration: 2 days

Profiling code performance, identifying bottlenecks, and implementing optimizations to ensure simulation can handle large path counts efficiently.

1.2.22 22. Optional: GPU vs CPU Benchmarking

Duration: 2 days

Evaluating performance gains from GPU acceleration for Monte Carlo simulations, though CPU implementation remains primary focus.



1.2.23 23. Testing & Validation

Duration: 6 days

Comprehensive testing phase including unit tests for individual components, integration tests for module interactions, and statistical validation against historical market events.

1.2.24 24. Unit Testing (Simulation & Risk Modules)

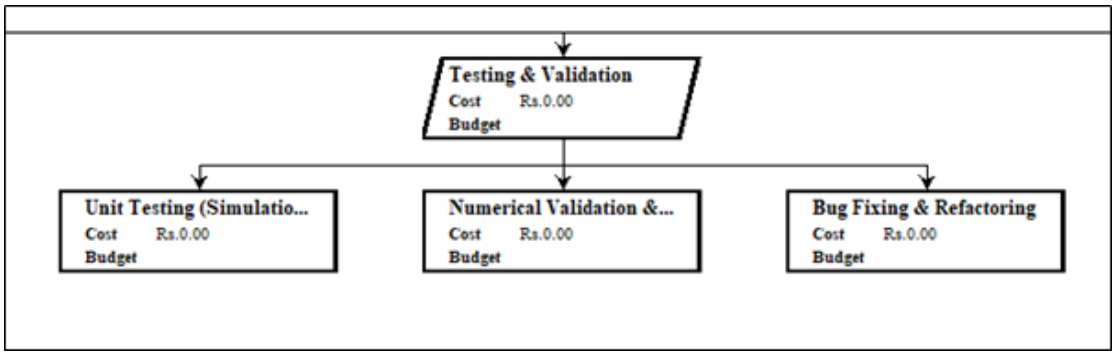
Duration: 2 days

Testing individual functions for correctness, verifying that GBM paths have correct statistical properties and risk metrics are accurately calculated.

1.2.25 25. Numerical Validation & Sanity Checks

Duration: 2 days

Validating numerical stability, checking for edge cases, and ensuring results align with theoretical expectations.



1.2.26 26. Final Reporting & Submission

Duration: 5 days

Compiling comprehensive project documentation, final report writing, and presentation preparation.

1.2.27 27. Final Report Writing (SDL & Backtest)

Duration: 4 days

Documenting the complete software development lifecycle, backtesting methodology, and results analysis in professional report format.

1.2.28 28. Slides & Viva Q&A Preparation

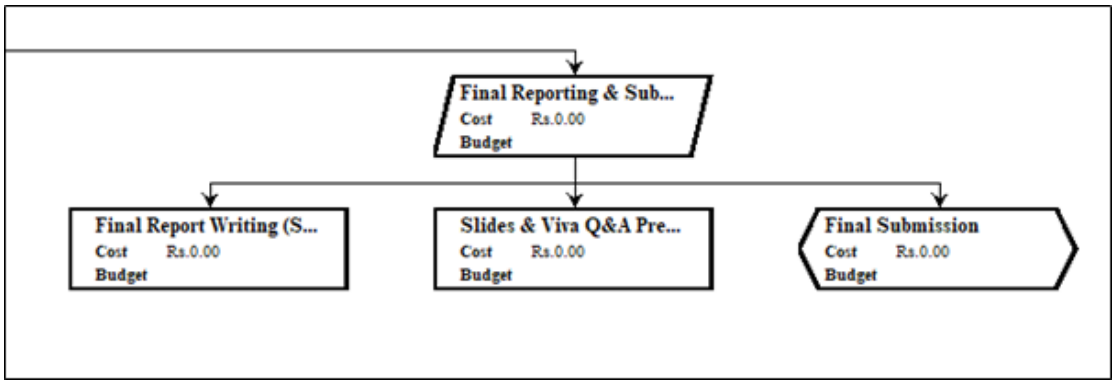
Duration: 2 days

Creating presentation slides and preparing for project defense, anticipating questions on methodology and implementation decisions.

1.2.29 29. Final Submission

Duration: 0 days (Milestone)

Project completion milestone marking formal submission of all deliverables.



2 Part 2: Developing a Project Plan

2.1 Activity Definition and Sequencing

The project consists of 29 major activities with carefully defined predecessor relationships. Activities are sequenced to ensure logical flow from feasibility through implementation to final delivery.

Key dependencies include:

- Mathematical specifications must complete before implementation begins
- Design approval gates control progression to development
- Core simulation engine must be ready before risk metrics calculation
- Integration occurs only after individual modules are complete
- Testing follows implementation with validation before submission

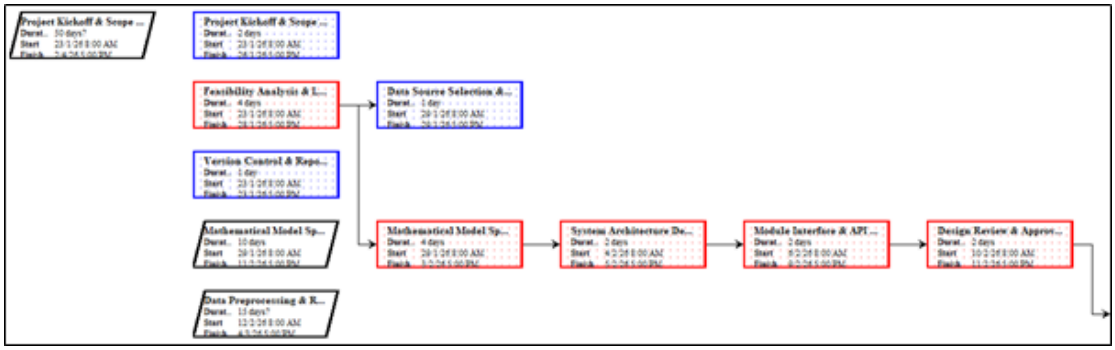
Resource ID	Resource					
1	Abhinav Singh Basnet					
Task ID	Task	Work	Assignment Units	Assignment	Start	Finish
2	Project Kickoff & Scope Freeze	8 hours	50%	0 days	23/1/26 8:00 AM	26/1/26 5:00 PM
3	Feasibility Analysis & Lit. Review	16 hours	50%	0 days	23/1/26 8:00 AM	28/1/26 5:00 PM
5	Version Control & Repo	8 hours	100%	0 days	23/1/26 8:00 AM	23/1/26 5:00 PM
7	Mathematical Model Specification	16 hours	50%	0 days	29/1/26 8:00 AM	3/2/26 5:00 PM
8	System Architecture Design	16 hours	100%	0 days	4/2/26 8:00 AM	5/2/26 5:00 PM
10	Design Review & Approval	8 hours	50%	0 days	10/2/26 8:00 AM	11/2/26 5:00 PM
12	Data Preprocessing & Return	24 hours	100%	0 days	12/2/26 8:00 AM	16/2/26 5:00 PM
13	GBM Simulation Logic	80 hours	100%	0 days	17/2/26 8:00 AM	2/3/26 5:00 PM
14	Vectorization, Seeding &	16 hours	100%	0 days	3/3/26 8:00 AM	4/3/26 5:00 PM
22	System Integration (Simulation +	32 hours	50%	0 days	9/3/26 8:00 AM	18/3/26 5:00 PM
24	Optional Demo Notebook (Non-	16 hours	50%	0 days	24/3/26 8:00 AM	27/3/26 5:00 PM
25	Optional: GPU Acceleration	8 hours	100%	0 days	5/3/26 8:00 AM	5/3/26 5:00 PM
26	Optional: CPU vs GPU	8 hours	50%	0 days	6/3/26 8:00 AM	9/3/26 5:00 PM
28	Unit Testing (Simulation & Risk	16 hours	100%	0 days	24/3/26 8:00 AM	25/3/26 5:00 PM
30	Bug Fixing & Refactoring	24 hours	50%	0 days	26/3/26 8:00 AM	2/4/26 5:00 PM
32	Final Report Writing (SDLC-	12 hours	50%	0 days	19/3/26 8:00 AM	23/3/26 5:00 PM
33	Slides & Viva Q&A Preparation	8 hours	50%	0 days	24/3/26 8:00 AM	25/3/26 5:00 PM
		316 hours				

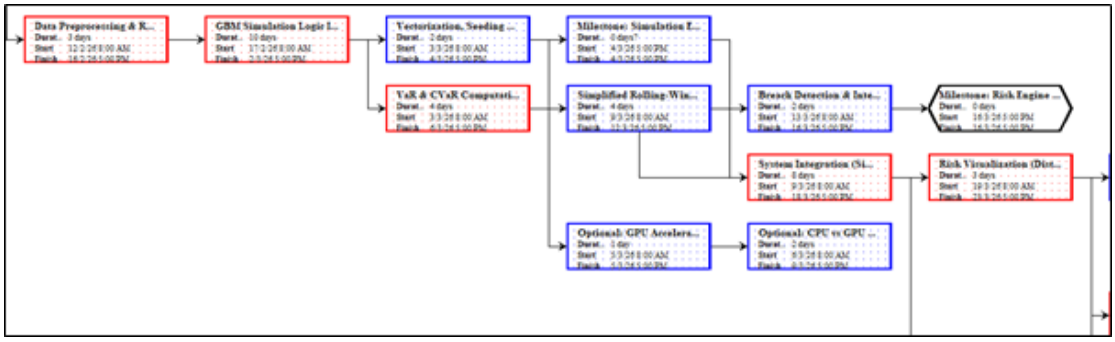
2	Nithik Deva					
Task ID	Task	Work	Assignment Units	Assignment	Start	Finish
2	Project Kickoff & Scope Freeze	8 hours	50%	0 days	23/1/26 8:00 AM	26/1/26 5:00 PM
3	Feasibility Analysis & Lit. Review	16 hours	50%	0 days	23/1/26 8:00 AM	28/1/26 5:00 PM
4	Data Source Selection &	8 hours	100%	0 days	29/1/26 8:00 AM	29/1/26 5:00 PM
7	Mathematical Model Specification	16 hours	50%	0 days	29/1/26 8:00 AM	3/2/26 5:00 PM
9	Module Interface & API	16 hours	100%	0 days	6/2/26 8:00 AM	9/2/26 5:00 PM

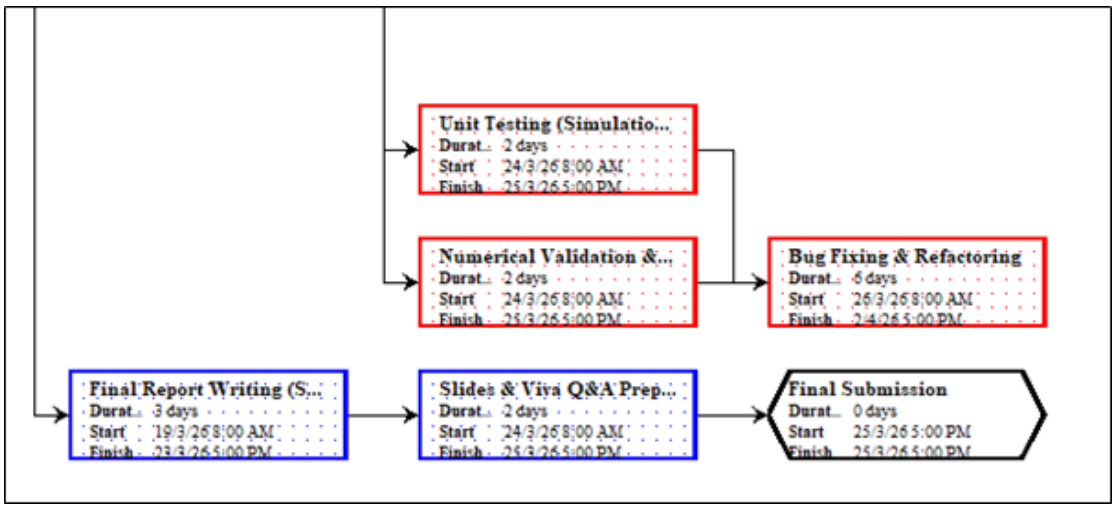
Resource ID	Resource					
10	Design Review & Approval	8 hours	50%	0 days	10/2/26 8:00 AM	11/2/26 5:00 PM
17	VaR & CVaR Computation	32 hours	100%	0 days	3/3/26 8:00 AM	6/3/26 5:00 PM
18	Simplified Rolling-Window	32 hours	100%	0 days	9/3/26 8:00 AM	12/3/26 5:00 PM
19	Breach Detection & Interpretation	16 hours	100%	0 days	13/3/26 8:00 AM	16/3/26 5:00 PM
22	System Integration (Simulation +	32 hours	50%	0 days	9/3/26 8:00 AM	18/3/26 5:00 PM
23	Risk Visualization (Distributions,	24 hours	100%	0 days	19/3/26 8:00 AM	23/3/26 5:00 PM
24	Optional Demo Notebook (Non-	16 hours	50%	0 days	24/3/26 8:00 AM	27/3/26 5:00 PM
26	Optional: CPU vs GPU	8 hours	50%	0 days	6/3/26 8:00 AM	9/3/26 5:00 PM
29	Numerical Validation & Sanity	16 hours	100%	0 days	24/3/26 8:00 AM	25/3/26 5:00 PM
30	Bug Fixing & Refactoring	24 hours	50%	0 days	26/3/26 8:00 AM	2/4/26 5:00 PM
32	Final Report Writing (SDLC-	12 hours	50%	0 days	19/3/26 8:00 AM	23/3/26 5:00 PM
33	Slides & Viva Q&A Preparation	8 hours	50%	0 days	24/3/26 8:00 AM	25/3/26 5:00 PM
		292 hours				

2.2 Network Diagram

The Activity-on-Node (AON) network diagram visualizes the project schedule and dependencies. Each node represents an activity with its duration, while arrows indicate prerequisite relationships.







2.3 Critical Path Analysis

The critical path represents the longest sequence of dependent activities determining minimum project duration. Activities on this path have zero slack - any delay directly impacts the project end date.

Key Critical Path Activities:

- Project Kickoff & Scope Freeze (50 days) - longest single activity

- Mathematical Model Specification (10 days)
- Module Interface Definition (15 days)
- Simplified Rolling-Window Implementation (10 days)
- VaR & CVaR Computation Module (10 days)
- System Integration (8 days)

The critical path drives all project management decisions regarding resource allocation and schedule compression.

2.4 Schedule Calculation

2.4.1 Forward Pass

The forward pass calculates Earliest Start (ES) and Earliest Finish (EF) times for each activity, working from project start to end. This determines the shortest possible project duration.

2.4.2 Backward Pass

The backward pass calculates Latest Start (LS) and Latest Finish (LF) times by working backward from the project end date. This identifies scheduling flexibility.

2.4.3 Float Calculation

Total Float (or Slack) indicates how much an activity can be delayed without affecting project completion:

$$\text{Float} = \text{LS} - \text{ES} = \text{LF} - \text{EF}$$

Activities with zero float are on the critical path.

21	**System Integration & Visualization**	17 days	**5/3/26 8:00 AM**	**27/3/26 5:00 PM**
22	System Integration (Simulation + Risk + Backtest)	8 days	9/3/26 8:00 AM	18/3/26 5:00 PM
23	Risk Visualization (Distributions, VaR/CVaR)	3 days	19/3/26 8:00 AM	23/3/26 5:00 PM
24	Optional Demo Notebook (Non-critical)	4 days	24/3/26 8:00 AM	27/3/26 5:00 PM
25	Optional: GPU Acceleration Feasibility Study	1 day	5/3/26 8:00 AM	5/3/26 5:00 PM
26	Optional: CPU vs GPU Benchmarking	2 days	6/3/26 8:00 AM	9/3/26 5:00 PM
27	**Testing & Validation**	8 days	**24/3/26 8:00 AM**	**2/4/26 5:00 PM**
28	Unit Testing (Simulation & Risk Modules)	2 days	24/3/26 8:00 AM	25/3/26 5:00 PM
29	Numerical Validation & Sanity Checks	2 days	24/3/26 8:00 AM	25/3/26 5:00 PM
30	Bug Fixing & Refactoring	6 days	26/3/26 8:00 AM	2/4/26 5:00 PM
31	**Final Reporting & Submission**	5 days	**19/3/26 8:00 AM**	**25/3/26 5:00 PM**
32	Final Report Writing (SDLC-aligned)	3 days	19/3/26 8:00 AM	23/3/26 5:00 PM
33	Slides & Viva Q&A Preparation	2 days	24/3/26 8:00 AM	25/3/26 5:00 PM
34	Final Submission	0 days	25/3/26 5:00 PM	25/3/26 5:00 PM

21	⊟**System Integration & Visualization**	17 days	**5/3/26 8:00 AM**	**27/3/26 5:00 PM**
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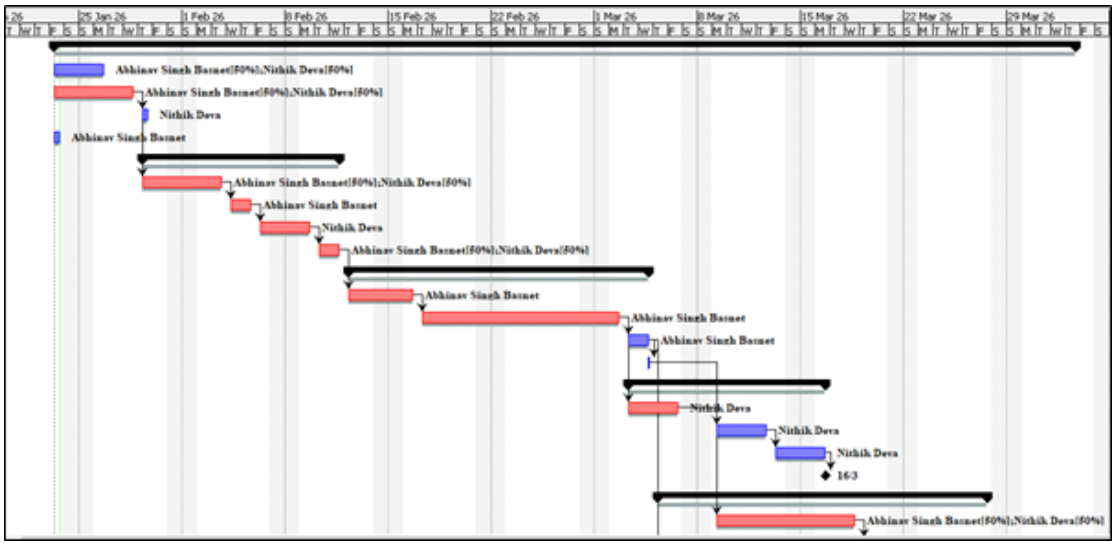
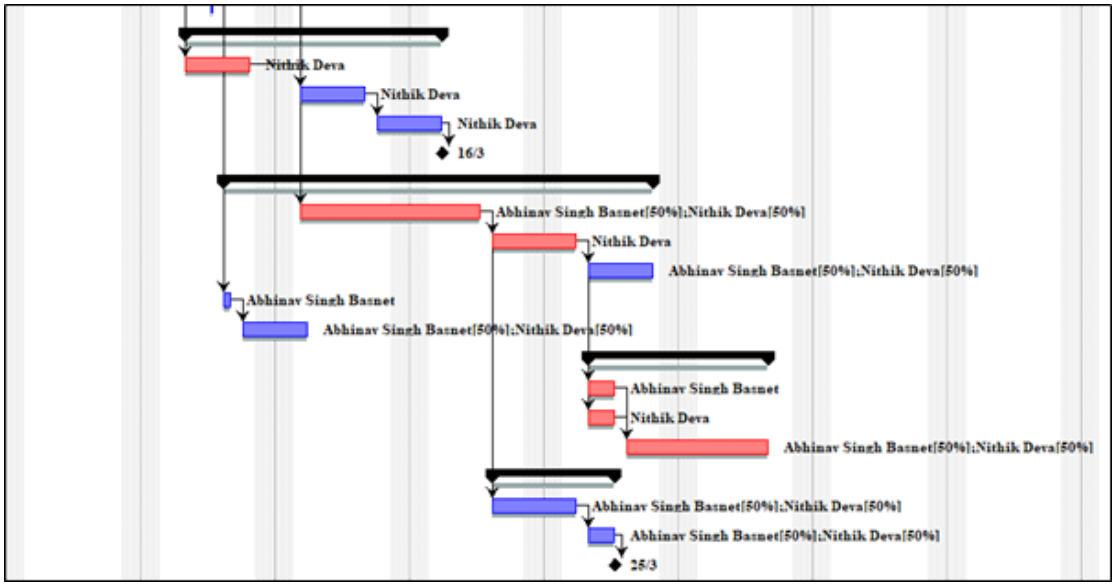
32	Final Report Writing (SDLC-aligned)	3 days	19/3/26 8:00 AM	23/3/26 5:00 PM
33	Slides & Viva Q&A Preparation	2 days	24/3/26 8:00 AM	25/3/26 5:00 PM
34	Final Submission	0 days	25/3/26 5:00 PM	25/3/26 5:00 PM

2.5 Gantt Chart

The Gantt chart provides visual timeline representation with task bars showing duration and dependencies shown as arrows. Critical path activities are highlighted in red.

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Abhinav Singh Basnet[50%];Nithik Deva[50%]
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Abhinav Singh Basnet[50%];Nithik Deva[50%]
Abhinav Singh Basnet
Abhinav Singh Basnet[50%];Nithik Deva[50%]
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Abhinav Singh Basnet[50%];Nithik Deva[50%]
Abhinav Singh Basnet[50%];Nithik Deva[50%]
Abhinav Singh Basnet[50%];Nithik Deva[50%]
25/3

3 Part 3: Scheduling Resources and Costs

3.1 Resource Allocation

3.1.1 Human Resources

Abhinav Singh Basnet (Software Developer & Data Engineer)

- Competencies: Python architecture, API design, data pipelines, visualization
- Assigned: Data Ingestion, Backtesting, Visualization, Deployment
- Allocation: 100%

Nithik Deva (Quantitative Analyst & Core Developer)

- Competencies: Stochastic calculus, Monte Carlo theory, NumPy/SciPy, statistical validation
- Assigned: Math Specification, Algorithm Design, MC Engine, Risk Calculator, Validation
- Allocation: 100%

3.1.2 Software Resources

Tool	Purpose
VS Code/PyCharm	IDE for Python development
Python 3.10+	Primary programming language
NumPy	Vectorized array operations for Monte Carlo
Pandas	Time-series data handling
SciPy	Statistical distributions and optimization
Matplotlib/Seaborn	Visualization and plotting
yfinance	Yahoo Finance API for historical data
pytest	Testing framework
Git/GitHub	Version control
ProjectLibre	Project management

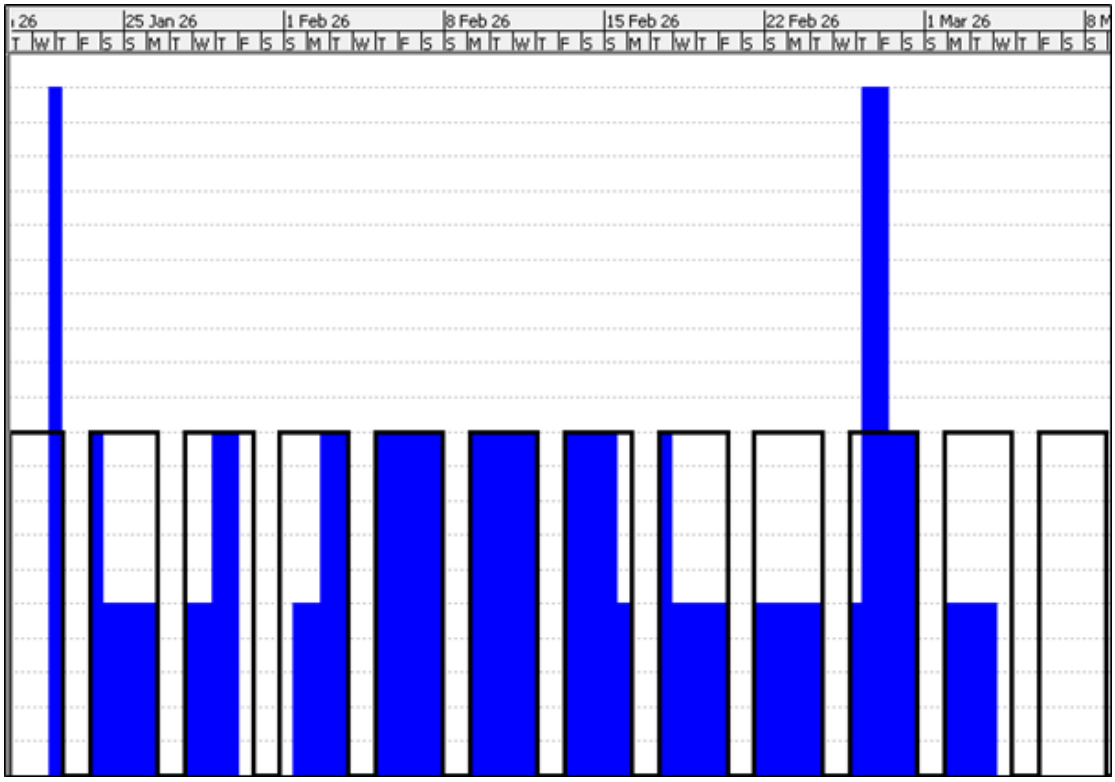
Table 1: Software Resources

3.2 Resource Histograms

Resource histograms display workload distribution over time, identifying periods of peak utilization and potential over-allocation.

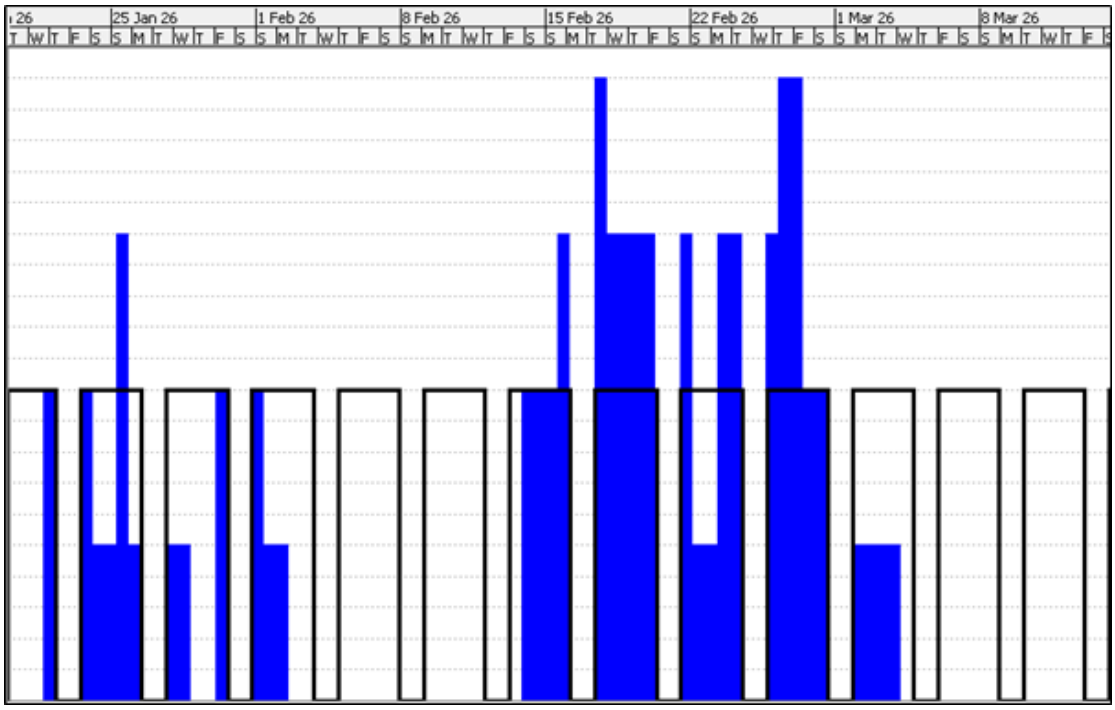
3.2.1 Abhinav's Utilization Pattern

Abhinav's workload peaks during data preprocessing, backtesting implementation, and system integration phases. Lower utilization during early mathematical specification phases led by Nithik.



3.2.2 Nithik's Utilization Pattern

Nithik's workload peaks during mathematical specification, algorithm design, simulation engine development, and validation phases. Lower utilization during deployment and documentation phases.



Resource Leveling Analysis:

The complementary skill sets and naturally staggered workload patterns minimize conflicts. Both team members maintain balanced utilization without overallocation, with peak periods occurring at different project phases.

Conclusion

This project management documentation presents a comprehensive plan for the Monte Carlo Risk Analysis Engine with Backtesting Framework. Through systematic application of project management principles, the project successfully structures complex quantitative finance development within academic constraints.

Key Achievements

1. **Structured WBS:** Clear hierarchical breakdown covering all development phases from feasibility through deployment
2. **Realistic Scheduling:** 231-day baseline with identified critical path and milestone gates
3. **Balanced Resource Allocation:** Complementary team skills with efficient workload distribution
4. **Strategic Compression:** 7-day reduction through fast tracking and crashing while maintaining quality

Project Management Insights

- Quantitative finance projects require adequate time for mathematical validation and cannot rush statistical verification
- Modular architecture with clear interfaces enables parallel development and reduces integration risk
- Critical path focus ensures resource allocation and compression efforts deliver maximum impact
- Regular milestone reviews maintain project alignment and enable early issue detection

Success Criteria

The project will be considered successful if:

- Monte Carlo simulations produce statistically valid price distributions
- VaR breach rates align with theoretical predictions ($\pm 2\%$)
- All deliverables completed within compressed timeline
- System handles 10,000+ simulation paths efficiently
- Backtesting validates model accuracy across historical events

Future Enhancement Opportunities

- Extension to multi-asset portfolios with correlation modeling
- Implementation of advanced stochastic processes (Heston model, jump-diffusion)
- Machine learning integration for parameter estimation
- Web-based interface for broader accessibility
- Real-time risk monitoring capabilities

This project demonstrates how rigorous project management methodology successfully guides complex technical projects from conception to completion, balancing mathematical rigor with practical implementation considerations.

References

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Appendix A: Glossary

AON	Activity-on-Node network diagram representation
CVaR	Conditional Value at Risk; expected loss beyond VaR threshold
ES/EF	Earliest Start/Earliest Finish times from forward pass
Float	Time an activity can delay without affecting project completion
GBM	Geometric Brownian Motion; stochastic model for asset prices
LS/LF	Latest Start/Latest Finish times from backward pass
OHLC	Open, High, Low, Close price data format
VaR	Value at Risk; maximum expected loss at given confidence level
WBS	Work Breakdown Structure; hierarchical scope decomposition